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Fraunhofer
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Master's Thesis

Multi-objective Optimisation of suction-cup and two finger grippers using todo: <Algorithims>

im Studiengang Automotive Systems
der FakultÃ¤t Fahrzeugtechnik
todo: <Semester>

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Eidesstattliche Erklärung

Hiermit versichere ich, die vorliegende Arbeit selbstständig und unter ausschließlicher Verwendung der angegebenen Literatur und Hilfsmittel erstellt zu haben.

Die Arbeit wurde bisher in gleicher oder ähnlicher Form keiner anderen Prüfungsbehörde vorgelegt und auch nicht veröffentlicht.

Esslingen, den 18. März 2026 ________________________

Unterschrift

Sperrvermerk

Die nachfolgende Master's Thesis enthält vertrauliche Daten der Fraunhofer IPA. Veröffentlichungen oder Vervielfältigungen dieser Arbeit – auch nur auszugsweise – sind ohne ausdrückliche Genehmigung der Fraunhofer IPA nicht gestattet. Diese Arbeit ist nur den Prüfern sowie den Mitgliedern des Prüfungsausschusses zugänglich zu machen.

Zitat

„Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea.“

Douglas Adams – The Hitchhikers Guide to the Galaxy

Vorwort

Dank an die Firma und die Firmenmitarbeiter, max. 1/2 Seite

Kurz-Zusammenfassung

„Aushängeschild"“ der Arbeit, max 1 Seite

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1 Milestone Timeline

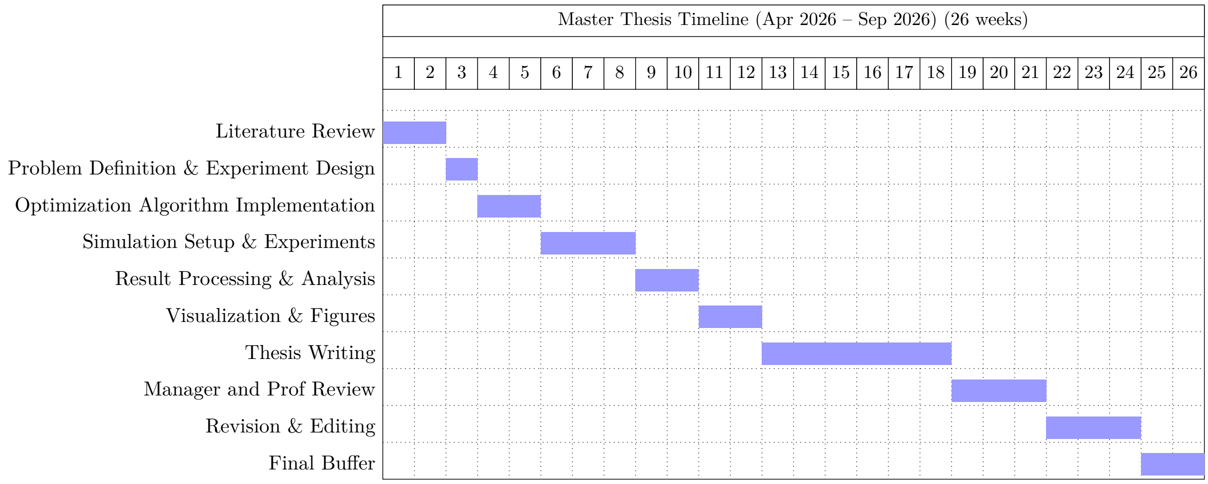

Abb. 1.1: Master thesis milestone plan

2 Einleitung

Die Gliederung des Hauptteils dieses Dokuments ist nur als Platzhalter gedacht. Im Folgenden sind zwei Vorschläge für Gliederungen dargestellt, die aber stets am konkreten Fall überprüft und in der Regel angepasst werden müssen.

Bitte beachten: Umfangreiche Programmlistings gehören nicht in die Arbeit, sondern auf eine beigefügte CDROM. Programmbeispiele können auszugsweise in der Arbeit gelistet werden, wenn sie zum Verständnis notwendig sind.

2.1 Literaturarbeit

1. Überblick (oder: Zusammenfassung, „Executive Summary", alles Wichtige für den „Manager" oder Schnellleser)
2. Fragestellung (oder: Ziele, Ausgangspunkt, Motivation)
3. Übersicht über den Stand der Wissenschaft und Technik (Beschreibung der Lösungsansätze, Beispiele etc. in einzelnen Abschnitten)
4. Bewertung der einzelnen untersuchten Ansätze, Beispiele etc., Identifikation von Defiziten
5. Synthese: Erstellung einer Gesamtschau, allgemeine Prinzipien, Beschreibung einer eigenen Sicht auf das Problem, evtl. auch eigene Vorschläge
6. Zusammenfassung (Erklärung des Nutzens), Ausblick

Anhang: eventuell recherchierte Texte, Produktbeschreibungen, etc.

2.2 System- und Softwareentwicklungen

1. Überblick (oder: Zusammenfassung, „Executive Summary", alles Wichtige für den „Manager" oder Schnellleser)
2. Problemstellung (oder: Ziele, Ausgangspunkt), Vorgesehener Benutzerkreis, Bedürfnisse der Benutzer
3. Stand der Technik (Wie wird das Problem bisher gelöst, wo sind die Defizite)
4. Gewählter Lösungsansatz (allgemeines Prinzip, welche Werkzeuge, z.B. Programmiersprachen werden verwendet)
5. Beschreibung der durchgeführten Arbeiten

6. Ergebnis (z.B. Screenshots mit Erläuterungen, Bedienanleitung)
7. Zusammenfassung (Erklärung des Nutzens), Ausblick

Anhang: evtl. (ausgewählte) Programmbeispiele.

3 Introduction

3.1 Motivation

Robotic manipulation plays a key role in modern automated production systems, particularly in logistics, assembly, and bin-picking applications. In such systems, the gripper forms the physical interface between the robot and the manipulated objects, directly influencing the reliability and efficiency of the manipulation process. Among the most commonly used gripper technologies are suction-cup grippers and two-finger mechanical grippers, each offering specific advantages depending on the properties of the objects to be handled.

Two-finger grippers provide versatile grasping capabilities for objects with varying geometries and are widely used in industrial manipulation tasks. Suction-cup grippers, on the other hand, are particularly effective for objects with smooth surfaces and enable fast and simple grasping operations with minimal mechanical complexity. However, determining suitable grasp configurations and gripper parameters for different tasks remains a challenging problem.

The performance of a gripper depends on multiple design and operational parameters such as finger geometry, gripping forces, actuator limits, and tool center point (TCP) positioning. These parameters are typically defined in configuration files describing the physical and functional properties of the gripper. For example, in the case of a two-finger gripper, parameters such as finger dimensions, actuator stroke, maximum gripping force, and TCP pose influence the resulting grasp quality. Similarly, suction-cup grippers depend on parameters including suction cup diameter, force limits, and allowable rotational deviation.

Traditional parameter tuning approaches often rely on heuristic rules or manual adjustments. Such approaches are time-consuming and do not guarantee optimal solutions, especially when multiple performance criteria must be considered simultaneously. In practical robotic applications, different objectives must be satisfied concurrently, including grasp stability, force limits, geometric feasibility, and manipulation efficiency.

Multi-objective optimization techniques provide a systematic approach to solving such problems by exploring the parameter space and identifying optimal trade-offs between competing objectives. Modern optimization algorithms such as Particle Swarm Optimization (PSO), Bayesian optimization methods, and Evolutionary Algorithms have demonstrated strong performance in solving complex, nonlinear optimization problems common in robotics.

Therefore, applying multi-objective optimization to the parameterization of suction-cup and two-finger grippers can significantly improve grasp planning and the overall performance of robotic manipulation systems.

3.2 Goal and Task of the Thesis

The goal of this thesis is to develop and evaluate a multi-objective optimization framework for determining optimal configurations of suction-cup and two-finger robotic grippers.

The grippers are defined through configuration files containing multiple adjustable parameters describing their mechanical properties, geometric dimensions, and operational limits. These parameters form the **decision variables** of the optimization problem. For the two-finger gripper, the configurable parameters include properties such as the number of fingers, actuator stroke limits, gripping effort, finger dimensions, and the position and orientation of the tool center point (TCP). In the case of the suction-cup gripper, relevant parameters include suction cup diameter, cup height, allowable rotation limits, and force limits such as coaxial, shear, and peeling forces.

The objective of the optimization process is to determine parameter combinations that result in improved grasp performance according to multiple criteria.

To achieve this goal, the following tasks are defined:

1. **Literature Review**

   Study existing research on robotic grasping, gripper design, and multi-objective optimization techniques applied in robotic manipulation and grasp planning.

2. **Modeling of Gripper Configurations**

   Analyze and model the configurable parameters of suction-cup and two-finger grippers based on their configuration files and identify the relevant decision variables for the optimization process.

3. **Definition of Objective Functions**

   Define multiple objective functions representing different aspects of grasp quality, such as grasp stability, force feasibility, collision avoidance, and geometric constraints.

4. **Implementation of Optimization Algorithms**

   Implement and apply several optimization algorithms, including Particle Swarm Optimization (PSO), Bayesian-based optimization methods, and Evolutionary Algorithms.

5. **Evaluation and Comparison**

   Evaluate the performance of the different optimization methods with respect to convergence behavior, computational efficiency, and the quality of the obtained solutions.

6. **Analysis of Optimization Results**

   Analyze the resulting Pareto-optimal solutions and identify suitable parameter configurations for both gripper types.

3.3 Approach

The grasp configuration problem is formulated as a multi-objective optimization problem in which several criteria relevant to grasp performance are optimized simultaneously. The adjustable parameters defined in the gripper configuration files serve as the decision variables of the optimization problem.

For the two-finger gripper, the decision variables include parameters related to the mechanical design and actuation system, such as actuator stroke limits, maximum gripping effort, finger geometry parameters (e.g., fingertip width and length), and the pose of the tool center point. For the suction-cup gripper, the decision variables include parameters such as suction cup diameter, cup height, rotation limits, and force limits including coaxial, shear, and peeling forces.

Based on these parameters, an optimization framework is established to evaluate different gripper configurations. Multiple objective functions are defined to quantify the quality of a given configuration according to criteria such as grasp feasibility, force limits, and geometric constraints.

To explore the search space efficiently, three different classes of optimization algorithms are applied:

- **Particle Swarm Optimization (PSO)**, a population-based optimization method inspired by collective behavior observed in natural systems.
- **Bayesian-based optimization methods**, which use probabilistic surrogate models to guide the search towards promising regions of the solution space.
- **Evolutionary Algorithms (EAs)**, which apply principles of natural evolution such as selection, mutation, and recombination to iteratively improve candidate solutions.

Each algorithm generates and evaluates candidate parameter configurations of the grippers. Since multiple objectives are considered simultaneously, the optimization process results in a set of **Pareto-optimal solutions** representing different trade-offs between the defined objectives.

The resulting solutions are analyzed and compared in order to evaluate the effectiveness of the different optimization approaches and to identify parameter configurations that improve the grasp performance of suction-cup and two-finger grippers.

4 Methodology

4.1 Methodology for Multi-Suction Cup Gripper Optimization

This section describes the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The goal is to determine optimal suction cup placements that maximize grasp performance across a set of workpieces.

4.1.1 Parameterization of the Gripper

The multi-suction gripper consists of three suction cups, each defined by its spatial position. Since the height (z-coordinate) and orientation of each suction cup are fixed, only the planar coordinates are considered as decision variables.

Thus, each candidate solution (individual) in the population is defined as:

$$\mathbf{p} = [(x_1, y_1), (x_2, y_2), (x_3, y_3)] \tag{4.1}$$

This configuration forms a triangular arrangement, which directly influences grasp stability and coverage. One assumption made here is that triangular configuration of the gripper gives a stable grip.

4.1.2 Initialization

An initial population of candidate solutions is generated randomly:

$$\mathcal{P} = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3, \dots\} \tag{4.2}$$

Each individual represents a unique triangular configuration of suction cup positions.

4.1.3 Workpiece Representation

The objects to be grasped (workpieces) are represented as point clouds. Each point in the point cloud corresponds to a potential grasping location.

4.1.4 Grasp Generation

For each individual **p** in the population, a subprocess is executed:

- For each point in the point cloud, a candidate grasp is generated using the tool center point (TCP) aligned with the surface point.
- The gripper is modeled as a rigid 3D structure with fixed height, forming a triangular prism based on the suction cup positions.

4.1.5 Collision Filtering

Each generated grasp is validated by performing collision detection between the gripper geometry and the workpiece.

- Grasps that result in collision are discarded.
- Only collision-free grasps are retained for further processing.

This step ensures that only physically feasible grasp configurations are considered.

4.1.6 Clustering of Valid Grasps

Due to the large number of valid grasp candidates, clustering is performed to group similar grasps:

- Only valid grasp points are clustered.
- Clustering is based on spatial proximity and similarity in orientation.

This reduces redundancy and allows efficient evaluation of grasp diversity.

4.1.7 Objective Functions

For each individual **p**, two primary objective functions are evaluated:

Triangle Area Maximization

The area of the triangle formed by the suction cups is computed as:

$$A(\mathbf{p}) = \frac{1}{2} \left|(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right| \tag{4.3}$$

The objective is to maximize this area to improve grasp stability, while ensuring it remains within feasible bounds of the workpiece surface.

Grasp Diversity Maximization

To ensure robustness across different grasp locations, a diversity metric is defined:

- The spatial distribution of clusters is evaluated.
- A normalized distance metric is computed between cluster centers.
- Orientation differences between grasps may also be incorporated.

The diversity score $D(\mathbf{p})$ is maximized to encourage wide coverage of the workpiece surface.

4.1.8 Optimization Loop

The evaluated objectives are passed to a multi-objective optimization algorithm:

$$\mathbf{f}(\mathbf{p}) = [-A(\mathbf{p}), -D(\mathbf{p})] \tag{4.4}$$

The optimization process follows an iterative loop:

1. Generate population $\mathcal{P}$
2. Evaluate each individual via grasp generation, collision filtering, and clustering
3. Compute objective values
4. Update population using optimization algorithm

This loop continues until a convergence criterion is met, such as:

- Maximum number of iterations
- Stabilization of Pareto front
- Desired objective threshold

4.1.9 Optimization Algorithms

The framework supports multiple optimization strategies:

- Particle Swarm Optimization (PSO)
- Bayesian Optimization
- Evolutionary Algorithms

Each algorithm generates new populations based on previous evaluations and explores the trade-off between triangle area and grasp diversity.

4.1.10 Output

The final result of the optimization is a set of Pareto-optimal configurations:

$$\mathcal{P}^* = \{\mathbf{p} \mid \mathbf{p} \text{ is Pareto-optimal}\} \tag{4.5}$$

These solutions represent optimal trade-offs between grasp stability and diversity, providing multiple candidate configurations for practical deployment.

5 Stand der Technik

Beschreibung des momentanen Standes der Technik und Wissenschaft. Hier muss auch klar definiert werden, in welchem Zusammenhang das Problem steht, d.h. der Systemkontext und der bekannte Lï¿½sungsraum ist klar zu beschreiben (vorhandene.Lï¿½sungen

5.1 Problembeschreibung

5.2 Existierende Lï¿½sungen

5.3 Literaturverweise

Verweise im Text: [1] und [Gun04].

6 Lï¿½sungsansatz

Hier kï¿½nnen je nach Art der Arbeit auch verschiedene Lï¿½sungsansï¿½tze beschrieben und diskutiert werden.

7 Durchgefï¿½hrte Arbeiten

Es muss klar erkennbar sein, auf was aufgesetzt wurde, d.h. schon vorhanden war, und was im Rahmen dieser Arbeit selbst entwickelt wurde.

8 Ergebnisse

Hier kann z.B. eine Bedienanleitung stehen, oder bei Analysen die Analyseergebnisse. „Neuigkeiten“ Messergebnisse

9 Zusammenfassung

Kurze Zusammenfassung und evtl. Beschreibung der Punkte, die zu einer kompletten, marktreifen Lï¿½sung noch fehlen.

10 Schluss

Ergebnis-Bewertung, Zusammenfassung und Ausblick

A Kapitel im Anhang

Alles was den Hauptteil unnötig vergrößert hätte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

Literaturverzeichnis

[1] Thomas Nonnenmacher, LaTeX Grundlagen - Setzen einer wissenschaftlichen Arbeit Skript, 2008,http://www.stz-softwaretechnik.de; *(Bei STZ Internetseite unter Publikationen - Skripte) [V. 2.0 26.02.08]*

[Gun04] Karsten Günther, LaTeX2 — Das umfassende Handbuch, Galileo Computing, 2004, http://www.galileocomputing.de/katalog/buecher/titel/gp/titelID-768; *1. Auflage*